

Joseph C. Slater, Ph.D., P.E.

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EDUCATION

<i>Institution</i>	<i>Concentration</i>	<i>Degree/Date</i>
State University of New York at Buffalo	Mechanical Engineering	Ph.D., 1993
State University of New York at Buffalo	Mechanical Engineering	M.S., 1992
State University of New York at Buffalo	Aerospace Engineering	B.S., 1989

PROFESSIONAL CREDENTIALS

Professional Engineer State Board of Registration for Professional Engineers and Surveyors, Ohio. Registration Number 66936, 2002.

PROFESSIONAL EXPERIENCE (ABBREVIATED)

<u>Position</u>	<u>Institution</u>	<u>Dates</u>
Dean- College of Engineering	Tennessee Technological University	7/19–present
Chair- Department of Mechanical and Materials Engineering	Wright State University	6/16–6/19
Director- Aerospace Professional Development Center	Wright State University	10/12–12/14
Interim Associate Vice President for Career and Workforce Development	Wright State University	1/14 – 3/15
Director, Master of Science Engineering, Aerospace Systems Engineering	Wright State University	10/12–6/19
Associate Dean for Defense Aerospace Studies (interim 8/11–11/11)	Wright State University	8/11–12/13

<u>Position</u>	<u>Institution</u>	<u>Dates</u>
Professor	Wright State University	9/08–6/19
Associate Professor	Wright State University	9/99–8/08
Assistant Professor	Wright State University	9/93–8/99
Postdoctoral Research Assistant	State University of New York at Buffalo	6/93–8/93
Research Assistant	State University of New York at Buffalo	6/89–5/93

PROFESSIONAL MEMBERSHIPS

- American Society of Mechanical Engineers, Fellow
- American Institute of Aeronautics and Astronautics, Associate Fellow
- American Society for Engineering Education, Member
- Tau Beta Pi

HONORS, AWARDS & RECOGNITIONS

<u>Organization</u>	<u>Award</u>	<u>Dates</u>
ASME	Fellow	2015
AIAA, Structural Dynamics, Mechanics, and Materials Organizing Committee	Outstanding service in organizing the SDM conference	2013
AIAA, Structural Dynamics, Mechanics, and Materials Organizing Committee	Outstanding service in organizing the SDM conference	2011
Dayton Affiliate Societies Council	Outstanding Engineers & Scientists Award, Education	2011
AIAA, Structural Dynamics, Mechanics, and Materials Organizing Committee	Outstanding service in assisting the SDM organizing committee in implementing ScholarOne for the first time	2010

<u>Organization</u>	<u>Award</u>	<u>Dates</u>
Wright Lab/AFRL	Summer Faculty Fellowship	1994, 1995, 1999, 2000, 2009, 2010
Wright State University, College of Engineering & Computer Science	Faculty Service Award	2006, 2010
Wright State University, College of Engineering & Computer Science	Outstanding Teaching Award Finalist	1999, 2004, 2007, 2008
DARPA	Certificate of Award for outstanding contributions to the ISAT program	8/2007
Wright State University, College of Engineering & Computer Science	Nominee for Wright State University Distinguished Professor of Service	2005
AIAA	Associate Fellow	2001
Phillips Lab	Summer Faculty Fellowship	1996
SUNY Buffalo	Presidential Fellowship	1989–1993
NASA Langley	Graduate Fellowship	1989–1992

PROFESSIONAL EXPERIENCE

Tennessee Technological University

Dean (2019-present), Professor (2019-present)

Leads a college of approximately **3,200 students** and **90 faculty**, with an annual non-research operating budget of roughly **\$26M**, **\$2.5M** in research center funding, and approximately **\$14M/year** in additional external research expenditures—the latter having quadrupled over the past 5 years.

- Created 3 new scholarships:
 - *Aspiring Engineering, Computing, & Technology Scholarship* for students who transform into superstars after enrolling.
 - *Engineering, Computing, & Technology Assistance Grant* providing emergency support to students who face dramatic life challenges impacting their ability to attend.
 - *College of Engineering Nonresidential Merit-Based Scholarship* supporting out-of-state students who are high-achievers yet not eligible for some merit-scholarships that require Tennessee residency.

- Supported the raising (with a wide-variety of level of contribution from one donor to another) of over **\$8M** towards the Ashraf Islam Engineering Building (including the largest ever cash donation to the university) by creating a strong vision for the building and building strong relationships with donors. After the building was awarded, rebuilt its vision from a collection of unit-owned spaces into a college-wide hub—with labs shared across departments and governed by cross-disciplinary faculty committees—designed to drive student belonging, recruitment, and a sense of home on campus; carried lead design responsibility for the building through that transformation, working with a team of three faculty who developed the building’s programming from that vision.
- Created the overall vision for and led the funding proposal for the Advanced Construction and Manufacturing Engineering Building (ACME), with three program areas developed by other individuals, delegating lead design responsibility to trusted colleagues while retaining overall project oversight.
- Invigorated curricular innovation resulting in multiple new minors and concentrations, with three being across multiple colleges.
- Oversaw the accreditation renewal efforts for seven undergraduate programs in 2020/2021.
- Oversaw the first accreditation of the joint General Engineering BS program with East Tennessee State University (ETSU) in 2022.
- Oversaw creation of the Nuclear Engineering BS program.
- Managed a **3%** budget cut in 2020 without resorting to terminations while actually growing faculty in multiple departments.
- Oversaw the hiring of **113** individuals into faculty or staff positions in the college including:
 - **56 faculty and 57 staff**
 - Marketing and Communications director
 - Associate Dean for Academic Affairs
 - Associate Dean for Research
 - Chair of Chemical Engineering
 - Chair of Manufacturing Engineering Technology
 - Chair of Electrical and Computer Engineering
 - Supported hire of Director and Associate Director of General Engineering program
 - Director of MS in Engineering Management
 - Supported the recruitment of multiple highly successful senior faculty

- Actively engaged multiple employers through advisory board and other direct relationships
- Expanded the college's K-12 and community outreach portfolio, adding a VEX Robotics high school competition, an annual Manufacturing Day, and hosting the state Quiz Bowl championship.
- Centralized college laboratory technicians—previously siloed one-per-department with uneven, mismatched skills—into a shared pool that moves between labs based on skill and central coordination while each retains primary responsibility for one lab, improving time-to-completion, work quality, lab appearance, and lab safety.
- Introduced normed quantitative performance metrics for use in assessing college, department, and individual performance for use in synergy with qualitative assessments.
- Supported numerous efforts directly tied to the university's "Rural Reimagined" grand challenge and numerous efforts in support of the university's vision and mission.
- Through the Budget Advisory Committee and other efforts, helped create strategies and practices that will ensure efforts align with university strategic goals and needs.
- Oversaw the first substantial increases in external expenditures in over eight years through successfully encouraging faculty to reengage with funded research. Expenditures on competitive grants and contracts rose **290%** over 6 years (from **\$3.5M to \$13.8M**).
- Redesigned academic advising from an ad hoc mix of faculty and staff advising into a centralized model of dedicated staff advisors paired with faculty mentoring, supporting growth in incoming freshman enrollment by **30%** over 6 years, an increase in retention from approximately **70% to over 80%**, a marked reduction in erroneous advising, and higher faculty productivity in teaching and research.
- Restructured college diversity efforts away from a single, siloed DEI officer position toward shared, distributed ownership—giving minority student organizations (SWE, SHPE, NSBE, and oSTEM) direct access to the dean and embedding diversity responsibility across the advising staff—growing the share of students identifying as women from **14.5% to 16%**, as Black from **3.5% to 4.1%**, and as students of color overall from **11.2% to 15.4%**, each outpacing regional trends.
- Supported the college becoming a National Academy of Engineers Grand Challenge College.
- Created a co-director position within the college's Student Success Center focused on holistic student support, and established a donor-funded student emergency fund, with a companion college food pantry established on the initiative of the college's Student Success Center team and development officer—together providing a discreet, single point of contact for students in crisis and preventing numerous cases of housing loss, food insecurity, or withdrawal from the university.

- Enacted the first data dissemination college-wide including:
 1. Enrollment numbers and trends
 2. External expenditure numbers and trends
 3. Initiation of the *Competitive Edge* report highlighting CoE achievements in research, scholarship, and service.
 4. Improved college funding decision-making through direct dissemination of amounts of Special Academic Fee (SAF, effectively differential tuition) funding and justifications to executive committee to be shared with faculty.
 5. Decentralized graduate student funding by directly transferring funding to departments collocating responsibility and authority

Wright State University

Chair (2016–2019), Professor (2008–2019), Associate Professor (1999–2008), Assistant Professor (1993–1999) – Department of Mechanical and Materials Engineering

- Served as coauthor with shared design responsibility for the \$37M, 90,000-square-foot Neuroscience Engineering Collaboration Building (completed 2015)—a four-story facility pairing neuroscience and engineering wings around a central atrium to foster translational research between engineering faculty, the Wright State Research Institute, and the Air Force Research Laboratory.
- Director (and founder) of MS in Aerospace Systems Engineering (2013–2019)
- Directed \$1.5M in research programs in the areas of modeling, control, and system identification of mechanical, aerodynamic, thermal, and electrical systems. Resulted in almost 70 technical articles and one book chapter.
- Wrote proposal and led development (Functional PI) of Ohio Mean's Interns and Co-ops I program resulting in \$1.3M in state funding. Led team to obtain employer commitment letters for over \$3.1M for internships in under one week.
- Led development of proposal for Ohio Mean's Interns and Co-ops II program resulting in \$800G (after planned reallocation of OMIC I funds from external expenditure to internal investment) in state funding. Led team to obtain commitment letters for over \$2.5M for internships in under one week.
- Supervised 18 graduate student (16 MS, 2 PhD) to completion as major advisor. A substantial number of students have become leaders in their industry or faculty.
- As Interim Associate Vice President, became the de facto leader of an after-hours, monthly cross-campus group of department chairs (spanning disciplines including English, Mathematics, and History) formed to share experience and support one another's success.
- Dayton Area's Affiliate Societies Council (that includes AIAA, ASME and other professional societies) recognized my contributions through the Outstanding Engineering and Scientist Award for Education in 2011.

- Recognized for service twice by Wright State's College of Engineering and Computer Science.
- Developed and coordinated what became Wright State's second-largest K-12 STEM outreach event, drawing as many as 800 students per year from hundreds of miles away over 15 years, run on an annual budget of roughly \$2,000 with about 20 staff and 50 student volunteers, with a deliberate focus on impact per dollar; a version of the event has since been adopted at another institution. Served as a STEM Fellow at the Montgomery County Regional STEM Center developing K12 STEM lesson plans meeting Ohio Department of Education guidelines (2008–2010).
- Received a Certificate of Award for Outstanding Contributions to the ISAT Program for work on the DARPA Innovative Satellite Antenna Program performing non-publishable research and analysis.
- Served as advisor to SWE for ten years personally advising many of the members, developing leadership skills, and providing and assisting with outreach opportunities targeting bringing young women into STEM fields.
- Served for several years as an *Associate Editor for Shock and Vibration* and the *International Journal of Modeling and Simulation*.
- Awarded two medals for service beyond supporting the AIAA SDM conference through transition to a new software system, overcoming and debugging the conference system, and serving on the AIAA conference software advisory committee, and a third for later service.
- Created the for-credit internship program within the College of Engineering and Computer Science (CECS), addressing a long observed need for both academically formalized experiential learning and greater support for connecting our students to industry. Developed a fiscally sustainable model for the intern program (see prior bullet) and programmed (with N. Klingbeil) the office to ensure greater interaction between intern office and students.

SCHOLARSHIP

Printed Scholarship

Journal Papers

1. Gillaugh, D. L., Kaszynski, A. A., Brown, J. M., Beck, J. A., and Slater, J. C., "Strain Gage Ramifications on Mistuning in As-Manufactured Models and Experimental Testing," *Journal of Engineering for Gas Turbines and Power*, Vol. 142, No. 5, 2020
2. Gillaugh, D. L., Kaszynski, A. A., Brown, J. M., Johnston, D. A., and Slater, J. C., "Accurate Strain Gauge Limits Through Geometry Mistuning Modeling," *Journal of Propulsion and*

Power, Vol. 34, No. 6, November 2018, pp. 1401–1408

3. Gillaugh, D. L., Kaszynski, A. A., Brown, J., Beck, J., and Slater, J. C., “Mistuning Evaluation Comparison via As-Manufactured Models, Traveling Wave Excitation, and Compressor Rigs,” *Journal of Engineering for Gas Turbines and Power*, Vol. 141, No. 6, 11 2018, pp. GTP–18–1617. Available from: <http://dx.doi.org/10.1115/1.4042079>
4. Beck, J. A., Brown, J. M., Kaszynski, A. A., Slater, J. C., and Cross, C. J., “Mistuned Response Prediction of Dual Flow-Path Integrally Bladed Rotors With Geometric Mistuning,” *Journal of Engineering for Gas Turbines and Power*, Vol. 137, No. 6, 2015. Available from: <http://dx.doi.org/10.1115/1.4028795>
5. Beck, J. A., Brown, J. M., Scott-Emuakpor, O. E., Cross, C. J., and Slater, J. C., “Dynamic Response Characteristics of Dual Flow-Path Integrally Bladed Rotors,” *Journal of Sound and Vibration*, Vol. 336, 2015, pp. 150 – 163. Available from: <http://dx.doi.org/10.1016/j.jsv.2014.10.011>
6. Scott-Emuakpor, O. E., Schwartz, J., George, T. J., Holycross, C., Cross, C. J., and Slater, J. C., “Bending Fatigue Life Characterisation of Direct Metal Laser Sintering Nickel Alloy 718,” *Fatigue & Fracture of Engineering Materials & Structures*, Vol. 38, No. 9, Feb 2015, pp. 1105–1117. Available from: <http://dx.doi.org/10.1111/ffe.12286>
7. Beck, J. A., Brown, J. M., Kaszynski, A. A., Cross, C. J., and Slater, J. C., “Geometric Mistuning Reduced-Order Models for Integrally Bladed Rotors With Mistuned Disk–Blade Boundaries,” *Journal of Turbomachinery*, Vol. 137, No. 7, 2015, pp. 11
8. Beck, J. A., Brown, J. M., Cross, C. J., and Slater, J. C., “Component-Mode Reduced-Order Models for Geometric Mistuning of Integrally Bladed Rotors,” *AIAA Journal*, Vol. 52, No. 7, July 2014, pp. 1345–1356
9. Beck, J. A., Brown, J. M., Cross, C. J., Slater, J. C., and Lamont, G. B., “Framework for Creating Digital Representations of Structural Components Using Computational Intelligence Techniques,” *AIAA Journal*, Vol. 52, No. 4, April 2014, pp. 855–866
10. Beck, J. A., Brown, J. M., Slater, J. C., and Cross, C. J., “Probabilistic Mistuning Assessment Using Nominal and Geometry Based Mistuning Methods,” *Journal of Turbomachinery-Transactions of the ASME*, Vol. 135, No. 5, September 2013
11. Shirayayev, O. V., Slater, J. C., and Brown, J., “Feasibility of Using Nonlinear Response Features for Crack Detection in Turbomachinery Components,” *AIAA Journal*, Vol. 51, No. 9, 2013, pp. 2290 – 2294. Available from: <http://dx.doi.org/10.2514/1.J052189>
12. Shirayayev, O. V. and Slater, J. C., “Detection of Fatigue Cracks Using Random Decrement Signatures,” *Structural Health Monitoring*, Vol. 9, No. 4, 2010, pp. 347–360
13. Shirayayev, O. V. and Slater, J. C., “Improved Structural Damage Identification Using Random Decrement Signatures: Application to FEM Data,” *Structural Control and Health Monitoring*, Vol. 15, No. 7, 2008, pp. 1006–1020

14. Balagangadhar, R. and Slater, J. C., "On the Convergence of Nonlinear Modes of a Finite Element Model," *Shock and Vibration*, Vol. 15, No. 6, 2008, pp. 655–664
15. Shiryayev, O. V., Page, S. M., Pettit, C. L., and Slater, J. C., "Parameter Estimation and Investigation of a Bolted Joint Model," *Journal of Sound and Vibration*, Vol. 307, No. 3-5, 2007, pp. 680–697
16. Shiryayev, O. V. and Slater, J. C., "Panel Flutter Model Identification Using the Minimum Model Error Method on the Forced Response Measurements," *ASME Journal of Vibration and Acoustics*, Vol. 128, No. 5, October 2006, pp. 635–645
17. Shiryayev, O. V. and Slater, J. C., "Aeroelastic System Identification Using the Minimum Model Error Method," *AIAA Journal of Guidance, Control, and Dynamics*, Vol. 29, No. 4, August 2006, pp. 936–943
18. Mortara, S. A., Slater, J. C., and Beran, P. S., "Analysis of Nonlinear Aeroelastic Panel Response Using Proper Orthogonal Decomposition," *Journal of Vibration and Acoustics-Transactions of the ASME*, Vol. 126, July 2004, pp. 416–421
19. Slater, J. C., Pettit, C. L., and Beran, P. S., "In-Situ Subspace Evaluation in Reduced Order Modelling," *Shock and Vibration*, Vol. 9, No. 3, 2002, pp. 105–122
20. Slater, J. C., Pettit, C. L., and Beran, P. S., "In-Situ Residual Tracking in Reduced Order Modelling," *Shock and Vibration*, Vol. 9, No. 3, 2002, pp. 105–121
21. Slater, J. C., Minkiewicz, G. R., and Blair, A. J., "Forced Response of Bladed Disk Assemblies – A Survey," *The Shock and Vibration Digest: Center for Intelligent Materials Systems and Structures*, Vol. 31, No. 1, 1999, pp. 17–24
22. Slater, J. C., "Application of the Nyquist Stability Criterion on the Nichols Chart," *Journal of Guidance Control and Dynamics*, Vol. 22, No. 2, March 1999, pp. 360–362
23. Wang, Y. and Slater, J. C., "A Comparison of Conventional and Impedance Methods for Modeling Piezoelectric Materials Actuation in Smart Structures," *Journal of Vibration and Acoustics*, Vol. 120, No. 3, 1998, pp. 685–688
24. Slater, J. C. and Inman, D. J., "On the Effect of Weak Non-Linearities on Linear Controllability and Observability Norms, an Invariant Manifold Approach," *Journal of Sound and Vibration*, Vol. 199, No. 3, January 1997, pp. 417–429
25. Slater, J. C., "A Numerical Method for Determining Nonlinear Normal Modes," *Nonlinear Dynamics*, Vol. 10, No. 1, 1996, pp. 19–30
26. Schulz, M. J., Thyagarajan, S. K., and Slater, J. C., "Inverse Dynamic Design Technique for Model Correction and Optimization," *AIAA Journal*, Vol. 33, No. 8, August 1995, pp. 1486–1491
27. Slater, J. C. and Inman, D. J., "Nonlinear Modal Control Method," *Journal of Guidance, Control, and Dynamics*, Vol. 18, No. 3, 1995, pp. 433–440

28. Banks, H. T., Wang, Y., Inman, D. J., and Slater, J. C., "Approximation and Parameter-Identification for Damped 2nd-Order Systems with Unbounded Input Operators," *Control-Theory and Advanced Technology*, Vol. 10, December 1994, pp. 873–892
29. Slater, J. C. and Inman, D. J., "Transfer Function Modeling of Damping Mechanisms in Distributed Parameter Models," *Mechanics Research Communications*, Vol. 20, No. 4, 1993, pp. 287–292

Books/Chapters

1. Cooley, P. E., Slater, J. C., and Shiryayev, O. V., *Simulating Fatigue Cracks in Healthy Beam Models for Improved Identification*, chap. 7: Fatigue of Materials II: Advances and Emergences in Understanding, Springer, 2012, pp. 87–101. Available from: <https://doi.org/10.1002/9781118533383.ch7>
2. Slater, J. C. and Agnes, G. S., *Nonlinear Modal Control Techniques and Applications in Structural Dynamic Systems*, Vol. 14, chap. 3: Structural Dynamic Systems, Computational Techniques and Optimization: Dynamic Analysis and Control Techniques, Gordon and Breach International Series in Engineering, Technology and Applied Science, 1999, pp. 107–168

Papers in Published Proceedings

1. Pardue, S. J., Yelamarthi, K., Chesson, T., Weathers, L. J., Bruce, J., and Slater, J. C., "Board 112: Creating an Institutional Culture of Empowering Faculty for Student-centered Learning through a Pilot Program," *2023 ASEE Annual Conference & Exposition*, ASEE, ASEE, 2023
2. Gillaugh, D. L., Kaszynski, A. A., Brown, J. M., Beck, J. A., and Slater, J. C., "Mistuning Evaluation Comparison via As-Manufactured Models, Traveling Wave Excitation, and Compressor Rigs," *ASME Turbo Expo 2018: Turbomachinery Technical Conference and Exposition*, No. GT2018-76888, 2018, p. V07CT35A039. Available from: <http://dx.doi.org/10.1115/GT2018-76888>
3. Slater, J. C. and Tidball, M. E., "Identification of Nonlinear Constitutive Properties of Damping Coatings," *2018 AIAA/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*, 2018, p. 0184
4. Tidball, M. E., Slater, J. C., Brown, J. M., Langley, B., and George, T. J., "Modeling and Analysis of Damping Performance of Hard Coatings in Turbomachinery," *58th AIAA/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*, 2017, p. 1979
5. Gillaugh, D. L., Kaszynski, A. A., Brown, J. M., Johnston, D. A., and Slater, J. C., "Accurate Strain Gage Limits Through Geometry Mistuning Modeling," *58th AIAA/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*, 2017, pp. AIAA 2017–0865. Available from: <https://doi.org/10.2514/6.2017-0865>

6. Park, S., Shin, S., Kim, Y., Matson, E. T., Lee, K., Kolodzy, P. J., Slater, J. C., Scherreik, M., Sam, M., Gallagher, J. C., Fox, B. R., and Hopmeier, M., "Combination of Radar and Audio Sensors for Identification of Rotor-Type Unmanned Aerial Vehicles (UAVs)," *2015 IEEE SENSORS - Proceedings*, Busan, Korea, Republic of, 2015. Available from: <http://dx.doi.org/10.1109/ICSENS.2015.7370533>
7. Cooley, P. E. and Slater, J. C., "Investigation of a Vibration-Based Damage Identification Technique for Breathing Fatigue Cracks," *56th AIAA/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*, 2015, pp. AIAA 2015-0693
8. Henry, E. B., Brown, J. M., and Slater, J. C., "A Fleet Risk Prediction Methodology for Mistuned IBRs using Geometric Mistuning Models," *17th AIAA Non-Deterministic Approaches Conference*, 2015, pp. AIAA 2015-1144
9. Cooley, P. E. and Slater, J. C., "Experimental Investigation of a Perturbation Model for the Nonlinear Response of a Fatigue Cracked Beam," *55th AIAA/ASME/ASCE/AHS/SC Structures, Structural Dynamics, and Materials Conference*, 2014. Available from: <http://dx.doi.org/10.2514/6.2014-0490>
10. Scott-Emuakpor, O., George, T. J., Beck, J., Schwartz, J., Holycross, C., Shen, M. H., and Slater, J., "Material Property Determination of Vibration Fatigued DMLS and Cold-Rolled Nickel Alloys," *Proceedings of the ASME Turbo Expo*, Vol. 7A, International Gas Turbine Institute, Dusseldorf, Germany, 2014. Available from: <http://dx.doi.org/10.1115/GT2014-26247>
11. Beck, J. A., Slater, J. C., Brown, J. M., and Cross, C. J., "Dynamic Response Characteristics of Dual Flow-Path Integrally Bladed Rotors," *52nd Aerospace Sciences Meeting*, 2014, pp. AIAA 2014-0098
12. Scott-Emuakpor, O., Schwartz, J., George, T. J., Holycross, C., and Slater, J., "Bending Fatigue Life Comparison Between Dmls and Cold-Rolled Nickel Alloy 718," *Machinery Failure Prevention Technology (MFPT) Society Annual Conference*, Virginia Beach, VA, May, 2014, pp. 20-22
13. Yelamarthi, K., Slater, J., Wu, Z., and Mawasha, P. R., "Engineering Management in an Interdisciplinary Senior Design Project," *Balkan Region Conference on Engineering and Business Education*, Vol. 1, De Gruyter Open, 2014, pp. 153-156
14. Beck, J. A., Brown, J. M., Cross, C. J., and Slater, J. C., "Geometric Mistuning Reduced Order Models for Integrally Bladed Rotors With Mistuned Disk-Blade Boundaries," *Proceedings of the ASME Turbo Expo*, Vol. 7 B, San Antonio, TX, United States, 2013. Available from: <http://dx.doi.org/10.1115/GT2013-94361>
15. Beck, J. A., Brown, J. M., Cross, C. J., and Slater, J. C., "Component Mode Reduced Order Models for Geometric Mistuning of Integrally Bladed Rotors," *AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference*, 2013

16. Cooley, P. E., Slater, J. C., and Shiryayev, O. V., "Improving Spectral Signature Profiles for Fatigue Crack Identification in Beams," *54th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*, 2013, p. 1498
17. Beck, J. A., Brown, J. M., Cross, C. J., and Slater, J. C., "Probabilistic Mistuning Assessment Using Nominal and Geometry Based Mistuning Methods," *Proceedings of the ASME Turbo Expo*, Vol. 7, Copenhagen, Denmark, 2012, pp. 1085 – 1097. Available from: <http://dx.doi.org/10.1115/GT2012-68533>
18. Cooley, P. E., Slater, J. C., and Shiryayev, O. V., "Fatigue Crack Modeling and Analysis in Beams," *53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference 20th AIAA/ASME/AHS Adaptive Structures Conference 14th AIAA*, 2012, p. 1874
19. Erford, M., Henry, E., Nay, J., and Slater, J., "Analysis of a Free-Fall Ballute Vehicle," *53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference 20th AIAA/ASME/AHS Adaptive Structures Conference 14th AIAA*, 2012, p. 1516
20. Slater, J. C. and Doman, D., "Proper Modes for Modeling Flapping Dynamics of Ornithopters," *53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference 20th AIAA/ASME/AHS Adaptive Structures Conference 14th AIAA*, 2012, p. 1981
21. Shiryayev, O., Gaerke, J., Cooley, P., and Slater, J., "Application of the Campbell Diagram Concept to Identification of Fatigue Cracks in Bladed Disk Assemblies," *52nd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference 19th AIAA/ASME/AHS Adaptive Structures Conference*, 2011, p. 1938. Available from: <https://doi.org/10.2514/6.2011-1938>
22. Shiryayev, O. and Slater, J., "Sensitivity Studies of Nonlinear Vibration Features For Detection of Cracks in Turbomachinery Components," *51st AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference 18th AIAA/ASME/AHS Adaptive Structures Conference 12th*, 2010, p. 3030
23. Snyder, J., Barnes, C., Rinderle, J., Shiryayev, O., and Slater, J. C., "Experimental Near-Space Free Fall Testing Systems," *51st AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference 18th AIAA/ASME/AHS Adaptive Structures Conference 12th*, 2010, p. 2982
24. Beck, J. A., Brown, J. M., and Slater, J. C., "Stochastic Mistuning Simulation of Integrally Bladed Rotors Using Nominal and Non-Nominal Component Mode Synthesis Methods," *4th Propulsion Safety and Affordable Readiness Conference*, March 2009, pp. 24–26
25. Meier, M., Shiryayev, O. V., and Slater, J. C., "Investigation of Candidate Features For Crack Detection in Fan and Turbine Blades and Disks," *50th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics & Materials Conference*, No. AIAA 2009-2671, 2009

26. Shirayayev, O. V. and Slater, J. C., "Supplemental Investigations on Structural Damage Detection Using Randomdec Signatures from Experimental Data," *50th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference 17th AIAA/ASME/AHS Adaptive Structures Conference*, 2009, p. 2330
27. Anisetti, A., Shirayayev, O., and Slater, J., "Non-Linear Shunting of Piezo Actuators for Vibration Suppression," *49th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference, 16th AIAA/ASME/AHS Adaptive Structures Conference, 10th AIAA Non-Deterministic Approaches Conference, 9th AIAA Gossamer Spacecraft Forum, 4th AIAA Multidisciplinary Design Optimization Specialists Conference*, 2008, p. 2237
28. Shirayayev, O. and Slater, J., "Structural Damage Identification Using Random Decrement Signatures from Experimental Data," *49th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference, 16th AIAA/ASME/AHS Adaptive Structures Conference, 10th AIAA Non-Deterministic Approaches Conference, 9th AIAA Gossamer Spacecraft Forum, 4th AIAA Multidisciplinary Design Optimization Specialists Conference*, 2008, p. 2163
29. Slater, J. C., "Making Math and Science Exciting Through Engineering Sport: the Wright State University Trebuchet Competition," *ASEE-NCS Conference*, February 2008, pp. 13.861.1 – 13.861.13. Available from: <https://peer.asee.org/4229>
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31. Nicholson, B., Page, S. M., Dong, H., and Slater, J. C., "Design of a Flapping Quad-Winged Micro Air Vehicle," *37th AIAA Fluid Dynamics Conference and Exhibit*, No. AIAA-2007-4337, AIAA, June 2007
32. Shirayayev, O. V. and Slater, J. C., "Improved Structural Damage Identification Using Random Decrement Signatures: Application to FEM Data," *48th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference*, April 2007
33. Mawasha, P. R., Yelamarthi, K., Wolff, J. M., Slater, J. C., and Wu, Z., "An Integrated Interdisciplinary Technology Project in Undergraduate Engineering Education," *Proceedings of 114th Annual ASEE Conference and Exposition*, 2007, pp. 24–27
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37. El-Ashry, M., Slater, J. C., Seethraman, G., and Young, H. D., "3D Displays based on Deformable Polydimethylsiloxane (PDMS) Lenticulars," *Materials Science and Technology*, ASM/TMS, September 2005
38. Pettit, C., Shiriyayev, O., Page, S. M., and Slater, J., "Parameter Identification and Investigation of a Bolted Joint Model," *46th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference*, No. AIAA-2005-2378, 2005
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43. Valevate, A., Slater, J. C., Cross, C. J., and George, T. J., "Semi-Active Vibration Control of Fan Blades," *9th National Turbine Engine High Cycle Fatigue Conference*, UTC, March 2004
44. Shiriyayev, O. V. and Slater, J. C., "Control of Resonant Fatigue Tests in the Existence of Bifurcations," *8th National Turbine Engine High Cycle Fatigue (HCF) Conference*, April 2003
45. Shiriyayev, O. V., Slater, J. C., Cross, C. J., and George, T. J., "Control of Resonant Fatigue Tests in the Existence of Bifurcations," *16th International Symposium on Air Breathing Engines*, No. ISABE-2003-1193, AIAA, Aug 2003, pp. 156–161
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55. Leo, D. J. and Slater, J. C., "Active/Passive Controller Synthesis Based on H Infinity Power Flow Techniques," *Smart Structures and Materials 1995: Mathematics and Control in Smart Structures*, Vol. 2442, International Society for Optics and Photonics, 1995, pp. 256–268
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61. Slater, J. C. and Inman, D. J., "Transfer Function Modeling of Damping Mechanisms in Distributed Parameter Models," *5th NASA Workshop on Distributed Parameter Modeling and Control of Flexible Space Structures*, edited by L. Taylor, Vol. 2, NASA, June 1992
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Conference Presentations (without paper)

1. Henry, E., and Slater, J.C., "Flight Dynamics System Identification of a Free Falling Bal-
lute," 38th AIAA Dayton-Cincinnati Aerospace Sciences Symposium, Dayton, OH, USA,
2013.
2. Cooley, P., Slater, J.C., and Shiryayev, O.V., "Improving Spectral Signature Profiles for
Fatigue Crack Identification in Beams," 38th AIAA Dayton-Cincinnati Aerospace Sci-
ences Symposium, Dayton, OH, USA, 2013.
3. Cooley, P., Slater, J.C., and Shiryayev, O.V., "Simulating Fatigue Cracks in Healthy Mod-
els for Improved Identification," 8th Annual Dayton Engineering Science Symposium,
Dayton, OH, USA, 2012. Best Paper: Solid Mechanics
4. Beck, J. A., Brown, J. M, and Slater, J. C., "Probabilistic Mistuning Assessment using
Frequency and Geometry Based Mistuning Methods, 6th Annual Propulsion - Safety
and Affordable Readiness Conference, Jacksonville, FL, USA, 2011.
5. Sharra, B, and Slater, J.C., "Zero Tension Release Mechanism," 5th Annual Dayton En-
gineering Science Symposium, October 2009.
6. Snyder, J, Slater, J.C., and Shiryayev, O.V., "High Altitude Balloon Free Fall Capsule Test,"
5th Annual Dayton Engineering Science Symposium, October 2009.
7. Beck, J. A., Brown, J. M, and Slater, J. C., "Stochastic Mistuning Simulation of Inte-
grally Bladed Rotors Using Component Mode Synthesis Methods," 34th AIAA Dayton-
Cincinnati Aerospace Sciences Symposium, Dayton, OH, USA, 2009.
8. Shiryayev, O.V., and Slater, J.C., "Supplementary Investigations on Detection of Fatigue
Cracks Using Randomdec Signatures from Experimental Data," 34th Dayton-Cincinnati
Aerospace Sciences Symposium, Dayton, OH, Feb. 2009.
9. Meier, M., Shiryayev, O.V., and Slater, J.C., "Investigation of Nonlinear Vibration Fea-
tures for Crack Detection in Fan and Turbine Blades and Disks," 34th Dayton-Cincinnati
Aerospace Sciences Symposium, Dayton, OH, Feb. 2009, **Best Technical Presentation
award.**
10. Shiryayev, O.V., and Slater, J.C., "Structural Health Monitoring Using Random Decre-
ment Signatures: Experimental Results," 33rd Dayton-Cincinnati Aerospace Sciences
Symposium, Dayton, OH, Feb. 2008.
11. Allen, C., and Slater, J.C., "Global Optimization of an Aircraft Thermal Management
Systems Using a Genetic Algorithm," 33rd Dayton-Cincinnati Aerospace Sciences Sym-
posium, Dayton, OH, Feb. 2008.
12. Nicholson, B., Page, S., Dong, H., and Slater, J.C., "Design of a Flapping Quad-Winged
Micro Air Vehicle," 37th AIAA Fluid Dynamics Conference and Exhibit, No. AIAA-2007-
4337, AIAA, June 2007.
13. Nicholson, B., Page, S, Dong, H., and Slater, J.C., "Wing Design And Testing For A Flap-
ping, Quad-Winged Micro Air Vehicle," 32nd Annual Dayton-Cincinnati Aerospace Sci-

ence Symposium, March, 2007, **Best Presentation- Aircraft Concepts and Design.**

14. Shirayayev, O.V., and Slater, J.C., "Structural Health Monitoring Using Randomdec Signatures: Application To FEM Data," 32nd Annual Dayton-Cincinnati Aerospace Science Symposium, March, 2007.
15. Shirayayev, O.V., and Slater, J.C., "Improved Structural Health Monitoring Using the Randomdec Signatures," 2nd Annual Dayton Engineering Sciences Symposium, presentation DESS06-0012, October 31, 2006.
16. Anisetti, A., Page, S.M., and Slater, J.C., "Suppressing Structural Response using Piezoelectric Patches," 2nd Annual Dayton Engineering Sciences Symposium, presentation DESS06-0058, October 31, 2006.
17. Page, S.M., Shirayayev, O.V., and Slater, J.C., "Investigation into Bolted Joint Dynamics," 2nd Annual Dayton Engineering Sciences Symposium, presentation DESS06-0070, October 31, 2006.
18. Esperanza, C., Page, S.M., and Slater, J.C., "Investigation of Vibrational Behavior Control Using Magneto-Rheological Fluids," 2nd Annual Dayton Engineering Sciences Symposium, presentation DESS06-0071, October 31, 2006.
19. Maddux, M., and Slater, J.C., "Using In-Situ Error Tracking For Mode Selection in Proper Orthogonal Decomposition Reduced Order Modeling," 2nd Annual Dayton Engineering Sciences Symposium, presentation DESS06-0102, October 31, 2006.
20. Shirayayev, O.V., Wolff, J.M., and Slater, J.C., "Power Extraction From Turbine Engines - Numerical Simulations," 2nd Annual Dayton Engineering Sciences Symposium, presentation DESS06-0032, October 31, 2006.
21. Meyer, A., Schrinner, A., Taphorn, A., and Slater, J.C., "Forced Response Testing System," 25th Annual Dayton-Cincinnati Aerospace Science Symposium, March 30, 2000.
22. Yelamarthi, K., Mawasha, P.R., Wolff, J.M., Slater, J.C., and Wu, Z., "Wright State University High Altitude Balloon Project," The Great Midwestern Space Grant Region's Small Balloon Conference, September 2006.
23. Shirayayev, O.V., Wolff, J.M. and Slater, J.C., "Turbine Engine Modeling," 31st Dayton-Cincinnati Aerospace Sciences Symposium, March, 2006.
24. Shirayayev, O.V. and Slater, J.C., "Sensor Effects on Identification Using the Minimum Model Error Algorithm," 31st Dayton-Cincinnati Aerospace Sciences Symposium, March, 2006.
25. Corbett, M., Williams, J., Walters, E., Lamm, P., Wolff, J.M. and Slater, J.C., "Analysis and Testing of Low Pressure Turbine Power Extraction Systems," 31st Dayton-Cincinnati Aerospace Sciences Symposium, March, 2006.
26. Page, S.M., Shirayayev, O.V., Slater, J.C., and Pettit, "Response Variability of a Bolted Joint," 31st Dayton-Cincinnati Aerospace Sciences Symposium, March, 2006.

27. El-Ashry, M., Slater, J.C., Young, D., and Seetharaman, G., "3-D Displays Based on Deformable Polydimethylsiloxane (PDMS) Lenticulars," *Materials Science & Technology 2005 Conference and Exhibition*, ASM/ACerS/AWS/AIST/TMS, Sept. 25–28, Pittsburgh, PA, 2005.
28. Shiryayev, O.V., and Page, S.M., and Slater, J.C. and Pettit, C.L., "Study of a Bolted Joint Model," 30th Dayton-Cincinnati Aerospace Sciences Symposium, February, 2005.
29. Page, S.M., Shiryayev, O.V., Slater, J.C., and Pettit, C., "Measurements and Modeling of Variability in the Dynamics of a Bolted Joint," *9th Joint Specialty Conference on Probabilistic Mechanics and Structural Reliability*, ASCE, Albuquerque, NM, July 26–28, 2004.
30. Shiryayev, O.V. and Slater, J.C., "Investigation of the Appropriate Excitation for Identification of Nonlinear Distributed Parameter Systems Using the Minimum Model Error Method," 29th Dayton-Cincinnati Aerospace Sciences Symposium, March, 2004.
31. Valevate, A.V., Slater, J.C. and Menart, J.A., "Semi-Active Vibration Control of Fan Blades," 29th Dayton-Cincinnati Aerospace Sciences Symposium, March, 2004.
32. Shiryayev, O.V., Slater, J.C., George, T., and Cross, C.J., "Control of Resonant Fatigue Tests in the Existence of Bifurcations," ISABE August 31–September 5, 2003, Cleveland, OH.
33. Shiryayev, O.V., and Slater, J.C., and Cross, C.J., "Control of Resonant Fatigue Tests in the Existence of Bifurcations," AIAA/ICAS International Air and Space Symposium and Exposition: The Next 100 Years, Poster presentation. 14–17 July 2003, Dayton, OH.
34. Valevate, A., and Slater, J.C., "Semi-Active Vibration Control of Fan Blades," AIAA/ICAS International Air and Space Symposium and Exposition: The Next 100 Years, Poster presentation. 14–17 July 2003, Dayton, OH.
35. Shiryayev, O.V., Slater, J.C., "Limit Cycle Oscillation Model Identification Using the Minimum Model Error Method," AIAA/ICAS International Air and Space Symposium and Exposition: The Next 100 Years, Poster presentation. 14–17 July 2003, Dayton, OH.
36. Shiryayev, O.V., Slater, J.C., and Cross, C.J., "Control of Resonant Fatigue Tests in the Existence of Bifurcations," High Cycle Fatigue Conference, 14–16 April 2003, Monterey CA.
37. Slater, J.C., Pettit, C. and Beran, P.S., "In-Situ Subspace Evaluation in Reduced Order Modeling," 42nd Structures, Structural Dynamics, and Materials Conference, Seattle, Washington, 2001.
38. Slater, J.C. and Sankarlal, S., "Blade Frequency Calculation For Tuned Aeroelastic Analysis," 25th Annual Dayton-Cincinnati Aerospace Science Symposium, March 30, 2000.
39. Gopinathan, A., Mortara, S., and Slater, J.C., "Limit Cycle Oscillation Model Identification Using the Minimum Model Error Method," 25th Annual Dayton-Cincinnati Aerospace Science Symposium, March 30, 2000.

40. Mortara, S., Slater, J.C., and Beran, P.S., "A Solution To The Nonlinear Flutter Problem Using Proper Orthogonal Decomposition," 25th Annual Dayton-Cincinnati Aerospace Science Symposium, March 30, 2000.
41. Wang, Y., and Slater, J.C., "Analysis of Piezoelectric Actuators Using a PDE Model," 1995 ASME Design Technical Conference.

Other Miscellaneous Presentations given since 2000

1. Slater, J.C., "Wright State University High Altitude Balloon Team: Our Lab is at 100,000 ft.," AIAA Dayton-Cincinnati chapter *Lunch 'n' Learn*, March 22, 2013.
2. Slater, J.C., "Insect-Inspired Mechanism for Ornithopter Flight Control", AFRL/VA, 2010.
3. Slater, J.C., Shirayayev, O.V., and Page, S.M., "Structural Damping and Joints - Guessing is No Longer an Option," Dayton chapter of the Society of Tribology and Lubrication Engineers, Jan. 11, 2006.
4. Slater, J.C., "In-Situ Subspace Evaluation in Reduced Order Modeling," AFRL/VA, 2000.

Other Public Visibility

1. "Wright State University Explores New Methods For Damage Detection in Turbomachinery Components," Insight (ABAQUS Magazine), 2010, No. 1.

Professional Conference Presentations by my Students (without advisor attribution)

1. Maddux, M., "Using In-Situ Error Tracking For Mode Selection in Proper Orthogonal Decomposition Reduced Order Modeling," 2nd Annual Dayton Engineering Sciences Symposium, October 2006.
2. Corbett, M., Williams, J., and Holtkamp, J., "Design of a High Altitude Balloon Payload," 2nd Annual Dayton Engineering Sciences Symposium, October 2006.
3. Holtkamp, J.C., Williams, J.M., and Corbett, M., "Design of High Altitude Balloon," 31st Dayton-Cincinnati Aerospace Sciences Symposium, March, 2006.
4. Corbett, M., "High Altitude Balloon Flight Path Prediction," 31st Dayton-Cincinnati Aerospace Sciences Symposium, March, 2006.

Published Software - technical

1. Slater, J.C., “Engineering Vibration Toolbox for Matlab 7.x,” 2006, for use with *Engineering Vibration*, Prentice Hall.
2. Slater, J.C., “Engineering Vibration Toolbox for Matlab 6.x,” 2002, for use with *Engineering Vibration*, Prentice Hall.
3. Slater, J.C., “Engineering Vibration Toolbox, 2nd Edition,” *Engineering Vibration*, Prentice Hall, New York, 1995.¹
4. Slater, J.C., and Leitner, J.A., *ctrb.m* and *obsv.m* of the *MATLAB Controls Toolbox*, 1996.
5. Slater, J.C., and Leo, D.J., “Vibration Toolbox,” *Engineering Vibration*, Prentice Hall, New York, 1993.

FUNDED GRANTS AND CONTRACTS

1. Slater, J.C., and Cooley, P.E., “Fatigue Crack Identification from Nonlinear Spectral Features in Bladed Disks,” DAGSI, \$27,000, 11/20/14 (continuation).
2. Sudkamp, T., Slater, J.C., and Miller, J., *Ohio Means Interns and Co-ops: Ohio Board of Regents*, \$800G + industry cost share of \$2.5M (TBD), 2014.
3. Slater, J.C., “OMIC - D’Angelo Technologies, LLC,” \$2,851, 7/13.²
4. Slater, J.C., “Turbine Engine Integrity Research,” Universal Technology Corporation, \$21,420, 1/13.
5. Sudkamp, T., Slater, J.C., and Miller, J., *Ohio Means Interns and Co-ops: Ohio Board of Regents*, \$1,304,631 + industry cost share of \$1.7M–\$3.2M (TBD), responsible for over 90% of internship commitments, 1/13–12/13.
6. Slater, J.C., and Cooley, P.E., “Fatigue Crack Identification from Nonlinear Spectral Features in Bladed Disks,” DAGSI, \$69,480, 6/12–6/13.
7. Fendley, R.D., and Slater, J.C., “WSARC Defense Aerospace Graduate Studies Institute Project,” State of Ohio, \$4,000,000, 7/11–6/13.
8. Slater, J.C., and Gaerke, J.P., “Application of the Campbell Design Concept to Identification of Fatigue Cracks in Bladed Disk Assemblies,” DAGSI, \$57,853, 6/10–5/11.
9. Slater, J.C., and Shiryayev, O.V., “Sensitivity Studies of Nonlinear Response for Crack Detection in Fan and Turbine Disks,” UTC/AFRL, \$50,000, 8/09–5/10.
10. Shiryayev, O.V., and Slater, J.C., “Vibration Tests of ITCN BCIT Card Cage Assembly,” ITCN Inc., \$2,021, 5/09–12/09.

¹Name was changed to resolve confusion with another toolbox with a similar name

²Tied to the Ohio Means Interns and Co-ops I program grant.

11. Mawasha, P.R., Slater, J.C., Wolff, J.M., and Wu, Z., "Enhancing Integrated Technology and Interdisciplinary Based Engineering Education Through the High Altitude Balloon (HAB) Experience," NSF CCLI, \$147,568, 9/08–8/11.
12. Shirayayev, O.V., and Slater, J.C., "Investigation of Candidate Features for Crack Detection in Fan and Turbine Disks," UTC, \$48,496, 8/08–11/08.
13. Meier, M., and Slater, J.C., "Crack Detection in Blades Using Random Decrement Signatures From Experimental Data," DAGSI, \$56,720, 3/08–3/09.
14. Mawasha, P.R., Wolff, J.M., and Slater, J.C., "Student BalloonSat Program," Ohio Space Grant Consortium, \$10,000, 4/06–4/07.
15. Slater, J.C., "Vibration Control of Turbo-Machinery Blade Vibration," UTC/AFRL, \$40,000, 1/07–11/07.
16. Wolff, J.M., and Slater, J.C., "Engine Health Monitoring via Dynamics Engine Model Failure Diagnostics," PCKA/AFRL, \$52,000, 7/13/06–7/12/07.
17. Slater, J.C., "Semi-Active Vibration Control of Fan Blades," UTC/AFRL, \$25,000, 5/1/06–6/30/07.
18. Slater, J.C., "ISAT Government SEIT Structures Reference Designs and Analytical Models," SAIC/DARPA, \$11,529, 9/13/04–6/30/06.
19. Slater, J.C., "Turbine Engine Modeling," PCKA, \$8,500, 1/12/05–2/28/06.
20. Slater, J.C., Menart, J.A., "Semi-Active Vibration Control of Fan Blades," UTC, \$25,000, 6/1/05–5/31/06.
21. Wolff, J.M., and Slater, J.C., "Power Extraction From a Gas Turbine Engine in Flight," PCKA, \$65,717, 4/1/05–6/30/06.
22. Slater, J.C., "ISAT Government SEIT Structures Reference Designs and Analytical Models," SAIC, \$12,957, 12/01/04–3/31/05.
23. Slater, J.C., "ISAT Government SEIT Structures Reference Designs and Analytical Models," SAIC, \$944, 9/13/04–9/30/05.
24. Slater, J.C., "ISAT Government SEIT Structures Reference Designs and Analytical Models," SAIC, \$18,289, 9/13/04–6/30/06.
25. Slater, J.C., "ISAT Government SEIT Structures Reference Designs and Analytical Models," SAIC, \$13,958, 9/01/04–3/31/05.
26. Wolff, J.M. and Slater, J.C., "Power Extraction From a Gas Turbine Engine in Flight," PCKA, \$70,000, 4/9/04–12/24/05.
27. Slater, J.C., and Wolff, J.M., "Parallel Reduced-Order, Modeling/In Situ Error Correction in CFD," NSF, \$25,740, 6/1/03–5/31/04.
28. Slater, J.C., "Quantifying Uncertainty in Structural Response," Anteon (AFRL/VA/AFOSR), \$156,825, 10/01/02–9/30/04.

29. Slater, J.C., "Validation and Enhancement to Wildcat - POD, AFRL/UTC," \$12,870, 6/03–12/03.
30. Slater, J.C., "Semi-Active Control of Fan Blades, Phase 3," AFRL/UTC, \$24,986, 1/04–12/04.
31. Slater, J.C., "Semi-Active Control of Fan Blades, Phase 2," AFRL/UTC, \$15,876, 6/03–12/03.
32. Slater, J.C., "Semi-Active Control of Fan Blades," AFRL/UTC, \$9,192, 3/03–6/03.
33. Slater, J.C., "Intergovernmental Personnel Agreement," AFIT, \$23,980, 6/02.
34. Slater, J.C., "Intergovernmental Personnel Agreement," AFIT, \$18,430, 9/02–12/02.
35. Slater, J.C., "P-POD: A Python-Based Generic Proper Orthogonal Decomposition Simulation Control Tool," UTC, \$36,405, 5/02.
36. Slater, J.C., "Development of a Novel Method for Evaluating Material Behavior Under Turbine Engine Operating Conditions," DAGSI, \$90,077, 7/1/00–6/30/02.
37. Slater, J.C., "Forced Response Enhancement to the Engine Structural Integrity Program Guidelines II," UTC, \$12,753, 8/11/00–2/15/01.
38. Slater, J.C., "Forced Response Enhancement to the Engine Structural Integrity Program," Universal Technology Corp., \$13,473, 8/99.
39. Slater, J.C., "Turbomachinery Dynamics Experimentation," Universal Technology Corp., \$14,256, 6/1/99.
40. Slater, J.C., "Electromechanical Engraving Device," Ohio Electronic Engravers: extension, \$5,980, 9/1/98.
41. Slater, J.C., "Field-of-View Research IPA," Naval Aeromedical Research Laboratory, \$10,000, 6/1/98.
42. Slater, J.C., "Experimental Study of Wave Propagation," Universal Technology Corp., \$5,000, 6/15/98.
43. Slater, J.C., "Electromechanical Engraving Device," Ohio Electronic Engravers, 3 months, \$7,535, 4/14/98.
44. Slater, J.C., Srinivasan, R., and Weiss, I., "Spring Manufacturer's Institute," SMI, Inc, \$10,080, 7 months, 1/13/98.
45. Slater, J.C., "Analysis of Stress Wave Propagation in Bladed Disk Assemblies," Universal Technology Corp., \$26,372, 6/17/97.
46. Slater, J.C., "A Design Strategy for Preventing High Cycle Fatigue," RDL (AFOSR), \$25,000, 10/25/95.
47. Lieh, J., and Slater, J.C., "Advanced 3D Camera Systems," Ohio Infrastructure Institute, \$80,920, 6/1/94.

OTHER FUNDED PROPOSALS

1. Narayanan, S. (PI), and Slater, J.C., “Neuroscience Engineering Collaboration Building (NEC),” Wright State University/State of Ohio, \$37,000,000, 2015.
2. Slater, J.C., Humphrey, E., Mohr, B., and Vondra, F., “Advanced Construction and Manufacturing Engineering Building (ACME),” State of Tennessee, \$89,600,000, 2022–2025.
3. Irwin, S., Yelamarthi, K., and Rao, M., “Nuclear Engineering Program Support,” U.S. Department of Education, \$3,000,000, 2023.

STUDENTS ADVISED (MS/PhD)

1. MacKenzie Tidball, “Identification of Nonlinear Constitutive Properties of Damping Coatings,” 2018, MS.
2. Emily Henry, “Stochastic Modeling of Geometric Mistuning and Application to Fleet Response Prediction,” 2014, MS.
3. Masoud Zarepoor, “Bulls-Eye Structure with a Sub-Wavelength Circular Aperture,” 2013, MS.
4. Joseph Beck, “Fundamental Understanding of Blisk Analytical Response,” 2013, Ph.D.
5. Joshua Mark, “Analytical and Experimental Vibration Analysis of Variable Update Rate Waveform Generation,” 2011, MS.
6. Joseph Beck, “Stochastic Mistuning Simulation of Integrally Bladed Rotors using Nominal and Non-Nominal Component Mode Synthesis Methods,” 2009, MS.
7. Anusha Aniseti, “Non-linear Shunting of Piezo-actuators for Vibration Suppression,” 2008, MS.
8. Chris Allen, “Global Optimization of an Aircraft Thermal Management System through Use of a Genetic Algorithm,” 2008, MS.
9. Oleg Shiryayev, “Improved Structural Health Monitoring Using Random Decrement Signatures,” 2008, Ph.D.
10. Jason Hansel, “The Influence of Thickness on the Complex Modulus of Air Plasma Sprayed Ceramic Blend Coatings,” 2008, MS.
11. Steve Page, “Investigation into the Behavior of Bolted Joints,” 2006, MS.
12. Mike Maddux, “Using In-Situ Error Tracking For Mode Selection in Proper Orthogonal Decomposition Reduced Order Modeling,” 2006, MS.
13. Brian Runyon, “The Influence Of Boundary Conditions And Aspect Ratio On Approximate Solutions For Constrained Layer Damping Treatments On Beams And Plates,”

2004, MS.

14. Anil Valevate, "Semi-Active Vibration Control of a Beam Using Embedded Magneto-Rheological Fluids," 2004 MS.
15. Oleg Shiryayev, "Investigation Of The Appropriate Excitation For Identification Of Non-linear Distributed Parameter Systems Using The Minimum Model Error Method," 2003, MS.
16. Mohamed Qureshi, "Robust Semi-Active Control of a Dry Friction Damper for a Cantilever Beam," 2001, MS.
17. Anil Gopinathan, "Robust Nonlinear System Identification Using Correlation Techniques," 1999, MS.
18. Andrew Blair, "A Design Strategy for Preventing High Cycle Fatigue by Minimizing Sensitivity of Bladed Disks to Mistuning," 1997, MS.
19. Ramesh Balagangadhar (Kurup), "On the Convergence of Nonlinear Modes Using the Finite Element Method," 1997, MS.

CONSULTING

1. Expert witness, Martin, McCarty, Wright & Roach, 2009.
2. Research Scientist, Paul C. Krause and Associates, 2005–2009.
3. Expert Witness, Wheelchair Tip Over Dynamics, Schuck Law Offices, 1999.
4. Deployment and Reorientation Modeling of the Innovative Space Antenna Technology program, SAIC, 2003.

SERVICE AND ACADEMIC OUTREACH

Associate Editor:

1. *Shock and Vibration* Journal, 2004–2011.
2. *International Journal of Modeling and Simulation*, 2004–2008.

Professional Society Committees

1. AIAA Structural Dynamics Technical Committee, 2004–present.

Major contributions:

- (a) AIAA SciTech Aerospace Design & Structures Technical Chair, 2019.
- (b) AIAA SciTech Aerospace Design & Structures Deputy Technical Chair, 2018.

- (c) Dynamic Specialists Conference Technical Discipline Chair (formerly called General Chair), 2016 SciTech
 - (d) Dynamic Specialists Conference Technical Chair (**split handling 187 abstracts**, reviews, and organized sessions), 2013 (SDM 54)
 - (e) Representative to SDM Program committee (**split handling 195 abstracts**, reviews, and organized sessions), 2011 (SDM 52)
 - (f) Hosted 2005 Fall Committee Meeting
 - (g) Chair: website committee, 2004–2010
 - (h) Member: Outreach committee– Converted award-winning **outreach DVD** for release Spring 2006 (1.0) and Spring 2007 (1.1) to web
 - (i) Chaired sessions (see below)
2. AIAA Dayton-Cincinnati Aerospace Sciences Symposium organizing committee, 2008, 2009, 2010.
 - (a) Publication chair, 2008.
 - (b) Publication co-chair, 2009, 2010.
 3. AIAA Conference Software Advisory Committee, 2011–2012.
 4. Member AIAA Gossamer Spacecraft Program Committee (2002–present)
Major contributions:
 - (a) **Website editor**
 - (b) Chaired sessions (see below)
 5. Montgomery County Regional STEM Center STEM Fellow.
Developed K12 STEM lesson plans tied to local research and industry needs and meeting Ohio Department of Education guidelines, 2008–2010.

Review Panels

- Proposal review
 1. NASA Large Aperture Research Program, panel **chair**, 2003.
 2. NSF Dynamic Systems & Controls, panel member, 2004.
 3. NASA New Millennium ST-9 Program, panel member, 2005.
- Program review
 1. Western Michigan University: STEP Advisory Board, 2012–present.

Reviewer:

1. Journals
 - (a) AIAA Journal

- (b) Shock and Vibration
- (c) AIAA Journal of Guidance, Control and Dynamics
- (d) ASME Journal of Vibration and Acoustics
- (e) International Journal of Modeling and Simulation
- (f) Journal of Engineering for Gas Turbines and Power
- (g) Journal of Intelligent Material Systems and Structures
- (h) Journal of Nonlinear Mechanics
- (i) AIAA Journal of Propulsion and Power
- (j) Journal of Vibration and Control
- (k) Mathematical and Computer Modeling
- (l) Mechanical Systems and Signal Processing
- (m) Nonlinear Dynamics

2. Books

- (a) Prentice Hall: Engineering Dynamics, 1st Edition, Tongue.
- (b) Prentice Hall: Engineering Mechanics: Statics, 1st Edition, Soutas-Little and Inman
- (c) Prentice Hall: Engineering Vibration, 2nd edition: Statics, Soutas-Little and Inman
- (d) Burkhauser: First Steps in $\text{\LaTeX}$, George Gratzer
- (e) McGraw Hill: Engineering Dynamics: Gray, Costanza and Plesha 1st draft
- (f) McGraw Hill: Engineering Statics: Gray, Costanza and Plesha 2nd draft
- (g) McGraw Hill: Engineering Dynamics: Gray, Costanza and Plesha 2nd draft

3. Conferences

- (a) International Gas Turbine Institute
- (b) AIAA SDM Conference, 40th, 41st, 42nd, 43rd, 45th, 46th, 47th, 48th, 50th
- (c) AIAA Gossamer Spacecraft Forum, 4th, 5th, 6th
- (d) AIAA Gossamer Spacecraft Forum Best Paper committee, 4th

Conference sessions chaired:

1. AIAA SDM-21 at 41st SDM Conference, 2000
2. AIAA SDM at 43rd SDM Conference, 2002
3. AIAA SDM at 44th SDM Conference, 2003
4. AIAA SDM at 45th SDM Conference, 2004
5. AIAA SDM-59 at 45th SDM Conference, 2004
6. AIAA GSF-6 at SDM Conference 4th Gossamer Spacecraft Forum, 2004
7. AIAA SDM-42 at 46th SDM Conference, 2005
8. AIAA SDM-59 at 46th SDM Conference, 2005
9. AIAA GSF-3 at 46th SDM Conference 6th Gossamer Spacecraft Forum, 2005
10. AIAA SDM-26 at 47th SDM Conference, 2006
11. AIAA GSF-8 at 47th SDM Conference 7th Gossamer Spacecraft Forum, 2006
12. AIAA SDM-9 at 48th SDM Conference, 2007
13. AIAA GSF-4 at 48th SDM Conference 8th Gossamer Spacecraft Forum, 2007
14. AIAA SDM-75 at 49th SDM Conference, 2008
15. AIAA SDM-34 at 50th SDM Conference, 2009
16. AIAA SDM-72 at 50th SDM Conference, 2009
17. MFPT 2009 Session 4A- Health Management for Maintenance and Decision Support, 2009

Committee service:*Department (Wright State University)*

1. Department Faculty Search Committees, 1994–1996, 1997–2002, 2007–2008, 2011–2013
2. Department Curriculum Committee, 1994–1996, 2003–2011 (chair 2006/2007–2011)
 - (a) Introduced program change to eliminate “soft” elective courses that were being misused by students
 - (b) Reviewed and approved Materials Minor program
 - (c) Participated in incorporation of design throughout the program as prescribed by ABET
 - (d) Oversaw largest overhaul of Mechanical and Materials Engineering programs in 14 years.
 - (e) Undergraduate semester transition committee, chair 2009–2019

- (f) Department internal advisory board, 2009–2010
- 3. Assistant to the Chair position search committee, chair, 2006 (2 times)
- 4. Student Recruitment position search committee, chair, 2006
- 5. Department Graduate Admissions Committee, 1997–2002
- 6. Honors and Awards Committee, 1997–1998

College (Wright State University)

- 1. College Library Committee, 1994–98 (Chair, 1996/97)
 - (a) Initiated the concept for a library copying service, adopted by the committee.
 - (b) Co-developed a web page for searching local resources focusing on the needs of CECS that streamlines the search process
- 2. College Curriculum Committee, 1996–1999, 2010 (Scribe 1996/7, Chair 1997–1999, 2010)
 - (a) Invited speakers from Math dept. to speak about courses they teach that are taken by our students (Calc. and DEs).
 - (b) Continued cooperation with Math dept. by initiating dept. by dept. evaluation of skills students need to acquire in Math courses.
 - (c) Oversaw two minors through the committee.
 - (d) Oversaw bulk of semester transition business
- 3. College PhD Program Coordinating Committee, 1996–2004
- 4. College PhD Program Student Affairs Committee, 2004–2019
 - (a) Assisted in the formulation of the Robotics and Controls Area program requirements.
- 5. College Teaching Committee, 1997–1999 (Scribe 1997/98)
- 6. College Steering Committee, 1999–2001, 2003–2009, Chair 2004–2005, 2005–2006, 2008–2009.
 - (a) Co-developed college bylaws with the committee
 - (b) Co-developed college awards procedures with the committee

University (Wright State University)

- 1. NTE faculty Bargaining Agreement Negotiation Committee (Administration)
- 2. University Undergraduate Curriculum and Policies Committee, 1996–1999, 2010–2011
- 3. University Legal Subcommittee of Campus-Wide Information System Advisory Committee, 1996–97

4. Graduate Council, 1996–1999
5. Faculty Senate, 1999–2001, *Engineering and Computer Science College Graduation Marshall, Fall 1999*
6. University Ad Hoc Committee for Undergraduate Academic Program Review, 2005–2006
7. University Ad Hoc General Education Semester Transition Committee, 2009

University (Tennessee Technological University)

1. Deans Council, standing member as Dean, 2019–present.
2. Budget Advisory Committee, standing member as Dean, 2019–present.
3. Academic Council, standing member as Dean, 2019–present.
4. Chaired the search committee for the university’s Director of Information Technology Services, 2022.

Advising

- Advisor of Society of Women Engineers, WSU student chapter, 2000–2009, 2010–2011.

Retention and Recruitment Activities

- Master of Ceremony for TEAMS bridge breaking competition (a competition for high school students as part of the Test of Engineering Aptitude and Mathematics), 1997, 1998.
- Designed the Wright State University Trebuchet competition, intended to be easier for students to test their designs at home or at school without expensive equipment; guidelines for the contest were made available on-line, 1999.
- Director, WSU Trebuchet Competition, 1999–2019. Integrated into the curriculum of over 20 school districts in Ohio.
- Taught the Computational Literacy portion of Wright State University’s College of Engineering and Computer Science Academic Advantage program, 1997, 1998, and re-designed the entire Computational Literacy portion of the course, 1998.

Open Source Software

1. **Engineering Vibration Toolbox**: A selection of 43 MATLAB and Octave codes for use in education of vibration phenomenon. Originally sponsored by the author of the corresponding text, it is now maintained pro-bono.
2. **Vibration Toolbox**: Educational module for illustrating vibration analysis and phenomenon. (Try on [mybinder](#))
3. **Mousai**: Harmonic Balance solvers and relevant tools to dynamic analysis to nonlinear dynamic systems.

4. [Array_to_LaTeX](#): Converts a NumPy/SciPy array or Pandas Numerical DataFrame to a $\LaTeX$ array or table using Python 3.x style (or alternatively) formatting of the result. (Try on [mybinder](#))
5. [VibrationTesting](#): A package for signal processing, modal analysis, and model reduction and model updating.
6. [Vitae](#): A Python module for building curriculum vitae and other documents from a Bib $\TeX$ file.
7. Other software (7 additional utilities not listed here).